

FM Data - Automation Platform

SRS Breakdown, Technical Solution & Delivery Approach

Power Platform Integrated Medallion Data Lakehouse Architecture

Document Control

Revision History

Ver.	Date	Author	Status	Summary of Changes
1.0	27-Aug-2026	Phuong Dang	Review	Initial breakdown of SRS into functional requirements, and technical architecture.

Purpose of this Document

This document decomposes the Facilities Management (FM) Software Requirements Specification (SRS) provided by RMIT Vietnam into structured functional and non-functional requirements, proposes a technical solution built on the Microsoft Power Platform with a Medallion (Bronze/Silver/Gold) data Lakehouse.

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See the companion workbook, FM_Product_Backlog_and_Sprint_Plan.xlsx, for the full Functional Requirements register, Product Backlog, Sprint Schedule, Resource Allocation Plan and RAID Log.

1. Executive Summary

Facilities Management (FM) operates across multiple campuses and building systems, generating data from Building Management Systems (BMS), Power Monitoring Expert (PME) / power-quality meters, offline meters, digital checklists, and a range of manual FM processes (continuous improvement requests, risk management, projects, CAPEX, OHS, training, vendor performance, and incident tracking). Today, this data and these processes are fragmented across systems, spreadsheets and point dashboards, making trending, comparison, and governance difficult.

This document proposes a single FM Data & Automation Platform built on the Microsoft Power Platform, with Microsoft Dataverse as Data Central — the one system of record for the whole platform — integrated with a Medallion (Bronze/Silver/Gold) Lakehouse in Microsoft Fabric, to: (1) automatically ingest utilities data (water, power, power quality) from BMS/PME into Dataverse; (2) automate FM workflows (CIWG, risk register, projects) with configurable approvals, notifications and escalation; (3) digitize checklists and offline meter readings via mobile-friendly forms; (4) consolidate FM KPIs (CAPEX, OHS, equipment/calibration, training, overtime, vendor, incidents) into a single, branded FM Portal; and (5) propose a phased roadmap for further automation.

Project Milestones

These four phases are the headline milestones for steering-committee/stakeholder reporting; Section 5.6 breaks each one down into its underlying two-week sprints.

Phase	Milestone	Sprint(s)	Date	Exit Criteria
1	Environment Access & Foundation	S1	07–18 Sep 2026	Power Platform (Dataverse as Data Central) and Fabric (OneLake) environments provisioned; Dataverse schema & Medallion data model approved; Link to Microsoft Fabric configured; BMS/PME/source-system access confirmed; Azure AD/Entra ID SSO configured; backlog refined.
2	Automated Utilities Data & Data Central	S2–S4	21 Sep–30 Oct 2026	Water and Power/PQM data flowing automatically into Dataverse (Data Central) and auto-syncing to Bronze → Silver → Gold via Link to Microsoft Fabric. Utilities Power BI dashboards (Direct Lake) live and accepted in UAT.
3	Functional Workflows & Power BI Hub	S5–S11	02 Nov 2026–05 Feb 2027	CIWG, risk-register and FM-projects workflows live with approvals → notifications → escalation; digital checklists and offline equipment capture live; all 7 FM KPI dashboards live; dashboards consolidated into the FM Portal / Power BI Hub with branding and less than 3s load.
4	Final Deliverable & User Training	S12	08–19 Feb 2027	End-to-end UAT signed off; user & admin documentation delivered; training sessions completed; post-go-live roadmap proposal delivered; production go-live and project closure.

How the SRS Maps to This Plan

No	Topic	Epic	SRS	Priority	Sprint(s)	Phase
1	Foundation: environment, data model, security	EPIC-0: Platform Foundation	—	Must	S1	Phase 1
2	Water utilities data integration	EPIC-1: Water Utilities Integration	1	Must	S2–S4	Phase 2
3	Power & power-quality data integration	EPIC-2: Power & PQM Integration	2	Must	S3–S4	Phase 2
4	Workflow automation for FM processes	EPIC-3: FM Workflow Automation	3	Must	S5–S6	Phase 3
5	Digital checklists & trending	EPIC-4: Digital Checklists & Trending	4	Must	S7	Phase 3
6	Offline equipment data capture	EPIC-5: Offline Equipment Data Capture	5	Must	S8	Phase 3
7	Integrated FM dashboards & KPIs	EPIC-6: Integrated FM Dashboards & KPIs	6	Must	S9–S10	Phase 3
8	Dashboard optimization & centralization	EPIC-7: FM Portal Optimization & Centralization	7	Must	S11	Phase 3
9	UAT, training, go-live, and maintenance	EPIC-8: Hardening, Go-Live & Maintain Support	—	Must	S12	Phase 4

2. Project Scope & Objectives

2.1 In Scope

- Design and implementation of online forms/templates: digital checklists, offline meter-reading forms, and FM workflow request forms.
- Automation solutions: Power Automate workflows, scheduled data jobs, and system-to-system data integrations (BMS, PME, offline sources).
- Data storage design and setup: a central SQL/Lakehouse environment (Bronze/Silver/Gold) and supporting data models.
- Visualization via web and mobile-friendly interfaces: Power BI dashboards and a centralized FM Portal.
- Custom components where Power Platform alone cannot reach a source system: Python and/or C# services for BMS/PME API or protocol gaps.
- A post-go-live roadmap proposal (near-term and longer-term) for further FM automation opportunities (Section 8).

2.2 Out of Scope (on this phase)

- Replacement of underlying BMS, PME or other OT/metering hardware — the platform integrates with what already exists.
- A detailed labor cost breakdown/commercial quotation — per stakeholder direction, this document focuses on scope, schedule and architecture; a costing exercise can be run as a separate, short follow-on activity once the backlog in the companion workbook is validated with operation FM (see Section 7).
- Non-FM enterprise systems (Finance/ERP, HR core systems) are beyond the data feeds needed for CAPEX, OT, and training KPIs.

2.3 Objectives

- Reduce manual data entry and spreadsheet-based reporting for utilities, checklists, and offline meters.
- Provide a single source of truth for FM data, governed through an auditable Bronze → Silver → Gold pipeline.
- Standardize and centralize FM workflows with visible status, due dates, and escalation.
- Give FM leadership, engineers, vendors and analysts role-appropriate, fast (<3s), branded dashboards from one FM Portal.
- Establish a platform and delivery cadence that can extend to further campuses, meters and FM domains without re-architecture.

3. Functional & Non-Functional Requirements

Requirements are grouped into nine epics (EPIC-0 foundation and hardening bookend the seven SRS-derived epics). Each requirement carries a MoSCoW priority.

The full register — cross-referenced to backlog user stories and sprints — lives in the companion workbook; the tables below are the authoritative requirement text.

3.1 EPIC-0 — Platform Foundation

ID	Requirement	Priority	SRS
FR-0.1	Provision a Power Platform environment (Dataverse as Data Central, DLP policies, solution-based ALM) and a Microsoft Fabric workspace (OneLake) dedicated to the FM platform.	Must	—
FR-0.2	Define the Dataverse table schema for utilities readings and FM-process entities, and the Medallion data model (Bronze/Silver/Gold) for the Fabric OneLake, including naming, retention and data-quality rules.	Must	—
FR-0.3	Integrate authentication/authorization with operation's identity provider (Azure AD / Entra ID) and define role-based access groups for FM Portal, Power BI and Power Apps.	Must	—
FR-0.4	Establish source-system connectivity and access (BMS, PME/equivalent, offline data owners) and confirm available APIs/protocols before building starts.	Must	—
FR-0.5	Configure Dataverse's "Link to Microsoft Fabric" so every relevant Dataverse table auto-syncs into Lakehouse, forming the Bronze layer without a separate ingestion pipeline.	Must	—

3.2 EPIC-1 — Water Utilities Data Integration

ID	Requirement	Priority	SRS
FR-1.1	Automatically collect water consumption data from BMS and/or related metering systems, at minimum hourly granularity, and write it into Dataverse (Data Central).	Must	1
FR-1.2	Water readings auto-sync from Dataverse into the Bronze layer unmodified (via Link to Microsoft Fabric), retaining source timestamp, meter ID and value.	Must	1
FR-1.3	Cleanse, validate (range checks, duplicate/missing-interval detection, unit consistency) and load water data into the Silver layer.	Must	1
FR-1.4	Aggregate Silver water data into Gold-layer tables supporting hourly/daily/monthly rollups by meter, building and campus.	Must	1
FR-1.5	Power BI dashboards visualize water consumption trends (hourly/daily/monthly) with comparison by campus, building and meter.	Must	1
FR-1.6	Dashboards support baseline and target comparison (e.g. % variance) where baseline/target values are configured.	Should	1
FR-1.7	Users can filter/drill down by date range, campus, building and meter across all water dashboards.	Must	1

3.3 EPIC-2 — Power & Power Quality Data Integration

ID	Requirement	Priority	SRS
FR-2.1	Automatically collect power consumption data (kWh, demand) from BMS and/or PME or equivalent systems and write it into Dataverse (Data Central).	Must	2
FR-2.2	Automatically collect power-quality metrics (voltage, THD and other configured PQM parameters) from PME/equivalent and write them into Dataverse (Data Central).	Must	2

ID	Requirement	Priority	SRS
FR-2.3	Power/PQM data auto-syncs from Dataverse into Bronze (via Link to Microsoft Fabric); cleansed and validated (incl. sensor-fault/outlier detection) in Silver; aggregated into Gold for long-term trending.	Must	2
FR-2.4	Gold-layer retention supports multi-year trending (configurable retention policy, e.g. minimum 3 years online).	Should	2
FR-2.5	Power BI dashboards display energy consumption (kWh, demand) and power-quality metrics (voltage, THD, etc.).	Must	2
FR-2.6	Dashboards support comparison across time ranges, feeders, buildings and campuses.	Must	2
FR-2.7	Configurable threshold-based alerts for power-quality anomalies (e.g. THD exceeding limit).	Could	2

3.4 EPIC-3 — Workflow Automation for FM Processes

ID	Requirement	Priority	SRS
FR-3.1	Provide a configurable workflow engine (Dataverse + Power Automate) supporting creation, modification, approval, rejection and closure of requests, reused across CIWG/continuous-improvement, risk register, and FM projects/initiatives.	Must	3
FR-3.2	Each workflow type supports configurable, multi-step approval routes and role-based approvers (e.g. by department, value threshold, risk severity).	Must	3
FR-3.3	Automated Email /MS Team notifications on key status changes: submitted, approved, rejected, overdue, escalated.	Must	3
FR-3.4	Track due dates per request/step and apply configurable escalation rules (e.g. reminder before due date, escalate to manager once overdue).	Must	3
FR-3.5	Retain full status history (approver, timestamp, comments) per request for audit purposes.	Must	3
FR-3.6	Users can view the status of their submitted requests and pending approvals via the FM Portal / Power Apps.	Must	3
FR-3.7	Risk register captures standard fields (description, likelihood, impact, owner, mitigation, review date) and produces a risk of heat-map view.	Should	3

3.5 EPIC-4 — Digital Checklists & Trending

ID	Requirement	Priority	SRS
FR-4.1	Provide configurable online/digital checklist forms (Power Apps) for operational and maintenance activities, accessible via web and mobile.	Must	4
FR-4.2	Forms support mandatory field validation, range checks (e.g. min/max for temperature/pressure) and unit selection.	Must	4
FR-4.3	Each checklist entry is automatically time-stamped and tagged with the submitting user ID.	Must	4
FR-4.4	Checklist data is stored centrally and available for trending of critical parameters (temperature, pressure, inspection results) in Power BI.	Must	4
FR-4.5	Checklist templates are configurable by asset type/location by an FM admin, without code changes.	Should	4
FR-4.6	Out-of-range entries are flagged for follow-up (visual indicator and/or notification to supervisor).	Should	4

3.6 EPIC-5 — Offline Equipment Data Capture

ID	Requirement	Priority	SRS
FR-5.1	Provide mobile-friendly online forms to key in manual readings from offline equipment (e.g. D1-CAMPUS water meters, NVL line meters, other offline meters).	Must	5
FR-5.2	Forms support offline data entry with local caching and sync-when-connected, for sites with unreliable connectivity.	Should	5
FR-5.3	Manually captured readings are stored centrally alongside automated readings, tagged with source = “manual”.	Must	5
FR-5.4	Dashboards display historical trends of manually captured readings and, where applicable, compare them with BMS/PME data for the same asset/location.	Must	5
FR-5.5	Configurable reminders for reading frequency (e.g. monthly reading due).	Could	5

3.7 EPIC-6 — Integrated FM Dashboards & KPIs

ID	Requirement	Priority	SRS
FR-6.1	CAPEX dashboard: plan vs. actual spend and project status, by project/campus.	Must	6
FR-6.2	OHS/safety dashboard: safety performance metrics and legal compliance status.	Must	6
FR-6.3	Equipment registers dashboard: register listing with calibration due/overdue and maintenance status.	Must	6
FR-6.4	Training & skills dashboard: completed training, expiring certifications and identified skill gaps by staff/role.	Must	6
FR-6.5	Overtime (OT) tracking dashboard/metrics by staff, department and period.	Must	6
FR-6.6	Vendor KPI dashboards: response time, quality and safety performance by vendor.	Must	6
FR-6.7	Incidents, near-misses and awards/recognition dashboard with trend and count views.	Must	6
FR-6.8	Each dashboard refreshes at a cadence appropriate to its source data (e.g. daily for HR/vendor data, near-real-time for utilities).	Should	6

3.8 EPIC-7 — FM Dashboard Optimization & Centralization

ID	Requirement	Priority	SRS
FR-7.1	Inventory and review existing FM dashboards/reports for consolidation, redundancy and gaps.	Must	7
FR-7.2	Re-publish/centralize all dashboards into a single FM Portal / landing page with role-based navigation.	Must	7
FR-7.3	Apply a consistent UI across dashboards: standard filters (campus, building, date range), operation branding (logo, colors, fonts) and consistent chart/table styling.	Must	7
FR-7.4	Main dashboards load within the agreed performance target (less than 3 seconds) under normal usage conditions.	Must	7
FR-7.5	Portal authentication/authorization aligns with operation's identity provider, with role-based access control.	Must	7

3.9 EPIC-8 — Hardening, Go-Live & Maintenance

ID	Requirement	Priority	SRS
FR-8.1	Conduct end-to-end UAT with stakeholders across all epics and resolve critical/high error defects prior to go-live.	Must	—
FR-8.2	Produce user and admin documentation (form/checklist configuration, workflow configuration, dashboard usage).	Must	—

ID	Requirement	Priority	SRS
FR-8.3	Deliver a post-go-live roadmap proposal for further FM automation opportunities (Section 8).	Must	—

3.10 Non-Functional Requirements

ID	Category	Requirement
NFR-1	Performance	Main dashboards load in less than 3 seconds under normal load; scheduled ETL/medallion pipelines complete within their defined batch window (e.g. nightly load complete before 06:00).
NFR-2	Availability	Platform available $\geq 99\%$ during operation business hours; planned maintenance windows communicated in advance.
NFR-3	Security	Role-based access control; encryption at rest and in transit; least-privilege service accounts for BMS/PME integration; alignment with IT security policy.
NFR-4	Data quality & governance	Documented validation rules and lineage from Bronze $\rightarrow$ Silver $\rightarrow$ Gold; immutable audit trail for manual entries and workflow actions.
NFR-5	Usability & accessibility	Mobile-responsive forms and dashboards; minimal training required for FM staff/technicians; (accessible color contrast and font sizing).
NFR-6	Scalability	Architecture accommodates additional campuses, buildings and meters without redesign of the medallion model or workflow engine.
NFR-7	Maintainability	Forms, checklists and workflows are configuration-driven (low-code) so FM/IT admins can adjust them without vendor engagement for routine changes.
NFR-8	Auditability	Full history retained for workflow requests and checklist submissions; who/when/what is always reconstructability.

4. Technical Solution Architecture

The solution is built on the Microsoft Power Platform with Microsoft Dataverse as Data Central — the single system of record every ingestion path writes into — integrated with a Medallion (Bronze/Silver/Gold) Lakehouse in Microsoft Fabric via Dataverse's native “Link to Microsoft Fabric” capability. This keeps one data model and one security model for the whole platform, while still giving utilities telemetry a proper analytical Lakehouse for long-term trending. Custom Python/C# services fill any gaps where a source system has no native low-code connector.

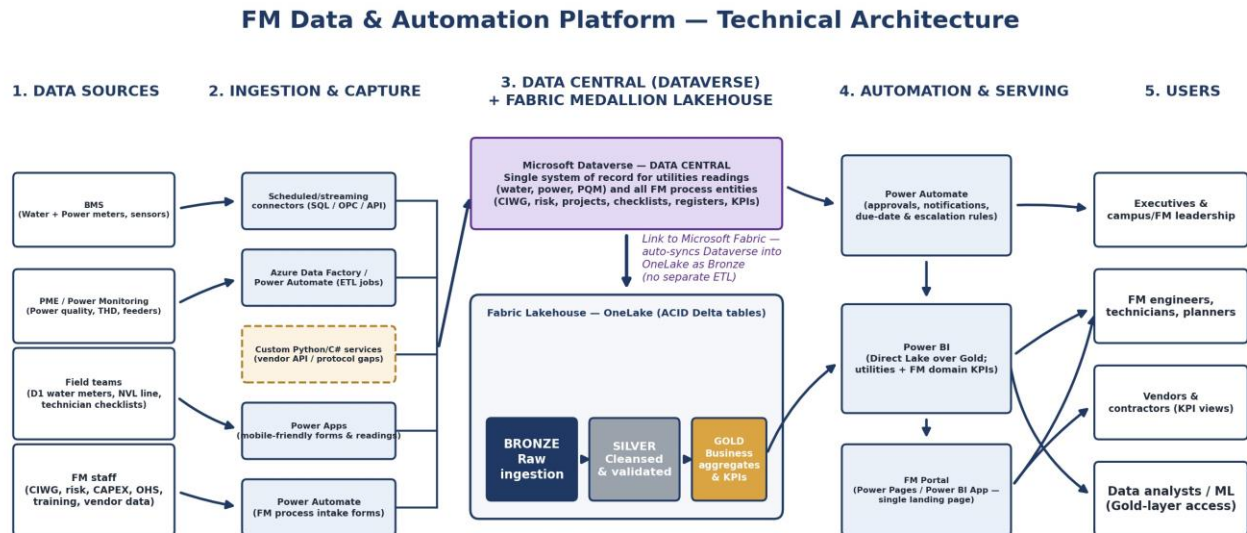


Figure 1 — FM Data & Automation Platform technical architecture: data sources → ingestion & capture → Dataverse (Data Central) + Fabric Medallion Lakehouse → automation & serving → users.

4.1 Data Central — Dataverse Integrated with the Medallion Lakehouse

- **Dataverse (Data Central):** every ingestion path — automated utilities feeds, Power Apps forms, Power Automate intake — writes into Dataverse tables first. It holds both the utilities readings (water, power, PQM) and every FM-process entity (CIWG requests, risk register, FM projects, checklist submissions, equipment register, training records, vendor KPI inputs), giving the whole platform one data model, one security-role model, and one place for Power Apps/Power Automate to read and write.
- **Link to Microsoft Fabric:** Dataverse's native, managed sync continuously mirrors every relevant table into OneLake as Delta Lake tables — with no separate ETL pipeline to build or maintain for that hop. This mirrored copy is the Bronze layer.
- **Silver (cleansed & validated):** Fabric Lakehouse notebooks/dataflows read Bronze and apply de-duplication, unit normalization, range/outlier checks, and gap detection (e.g. missing meter intervals), producing a trustworthy, query-ready layer.
- **Gold (business aggregates):** curated, campus/building/meter-level rollups (hourly/daily/monthly), baseline/target reference tables, and KPI calculations, stored as Delta tables in OneLake.
- **Serving back to Power BI and Power Apps:** Power BI reads Gold directly via Direct Lake mode — near-instant and always current, without a separate import/refresh schedule. Where a Power App needs a computed KPI (e.g. a rolled-up consumption figure on a form), Gold data is surfaced back into Dataverse via virtual tables, so the app still only ever talks to Dataverse.
- **Why this shape:** it satisfies “Dataverse is Data Central” literally — one system of record and one integration point for every source — while still giving high-volume utilities telemetry a real analytical lakehouse (ACID Delta tables, time-travel, cheap long-term storage) rather than forcing years of hourly meter readings to live only inside Dataverse's transactional tables. See the capacity/cost caveat in Section 7, risk R6.

4.2 Power Platform Components

Component	Role in this solution
Microsoft Dataverse	Data Central: single system of record for utilities readings and every FM-process/checklist/register entity; enforces validation, security roles and business rules; integrates natively with Power Apps/Automate.
Microsoft Fabric (OneLake)	Hosts the Medallion Lakehouse (Bronze/Silver/Gold) as Delta tables. Bronze is populated automatically by Dataverse's "Link to Microsoft Fabric"; Silver/Gold are built with Fabric notebooks/dataflows.
Power Apps	Mobile-friendly forms for digital checklists, offline meter readings, and FM-process request intake — model-driven/canvas apps over Dataverse.
Power Automate	Scheduled/event-driven jobs that pull BMS/PME data into Dataverse, approval routing, notifications (email/Teams), due-date and escalation logic.
Power BI	Semantic models and dashboards using Direct Lake mode over the Fabric Gold layer, plus live Dataverse queries for operational/workflow status; row-level security by campus/building.
Power Pages (FM Portal)	Single, branded landing page consolidating dashboards, forms and workflow status behind role-based, SO-integrated access; alternative is a curated Power BI App workspace if a full external-facing portal is not required.
Azure Data Factory (as needed)	Only for source systems that cannot reasonably write into Dataverse directly (e.g. a very high-frequency historian) — lands data into Dataverse via its Web API, or straight into a Fabric pipeline feeding the lakehouse, per the exception noted in 4.3.
Python / C# services	Small, targeted services for BMS/PME vendor APIs or protocols without a native connector (e.g. OPC UA, proprietary REST/SOAP APIs, file drops); write into Dataverse via its Web API, keeping Dataverse the single entry point.

4.3 Integration Approach

- BMS and PME/PQM: where a native or ODBC/SQL/OPC connector exists, a scheduled Power Automate job (or a lightweight Data Factory pipeline) reads the BMS/PME database or historian and writes readings into Dataverse tables. Where no connector exists, a small Python or C# service polls the vendor API/protocol and writes into Dataverse via its Web API — either way, Dataverse is the landing point, not a separate database.
- Offline meters & checklists: captured directly into Dataverse via Power Apps, so no separate ingestion step is required — these already start as structured, validated Dataverse records.
- FM systems of record (CAPEX/finance extracts, HR/training data, vendor records): connected via scheduled exports/APIs into Dataverse, or entered/maintained directly in Dataverse where client prefers to run them inside the platform rather than a source system.
- Exception for extreme-volume telemetry: if a specific meter/feed turns out to be too high-frequency for Dataverse to hold economically (see risk R6), that one feed can bypass Dataverse and land straight into the Fabric lakehouse (Bronze) instead, with only its aggregated/rolled-up figures written back into Dataverse for Power Apps visibility. This is an exception path, not the default, and would be confirmed during Sprint 1/2 once real data volumes are known.

4.4 Security & Governance

- Azure AD / Entra ID single sign-on for the FM Portal, Power Apps and Power BI; role-based security groups aligned to FM org structure (technicians, engineers, managers, executives, vendors), enforced once at the Dataverse layer and inherited everywhere else.

- Power Platform Data Loss Prevention (DLP) policies restrict which connectors can be used together, protecting BMS/PME credentials and Dataverse data.
- Application Lifecycle Management (ALM) via Dataverse solutions across Dev → Test → Prod environments, with source control for Power Automate/Apps components and CI/CD where practical.
- Data classification and retention policy defined per layer: Dataverse holds a shorter operational retention window for high-volume telemetry tables (managed via Dataverse's short-term retention policies) while full history is retained in OneLake through the Fabric Link (per NFR-4 for multi-year trending).

4.5 FM Portal & UX Standards

- One landing page (Power Pages or a curated Power BI App) replacing the current scattered dashboards/reports, per SRS no.7.
- Consistent global filters — campus, building, date range — applied across embedded dashboards wherever the data model supports it.
- Client branding (logo, color palette, typography) applied via a shared Power BI theme file and Power Pages template.
- Performance budget of less than 3 seconds for main dashboards, achieved through Gold-layer pre-aggregation, import/direct query mode selection per dashboard, and incremental refresh.

5. Agile-Scrum Delivery Approach

5.1 Methodology

Delivery follows Scrum with two-week sprints. Given the breadth of the SRS relative to a lean delivery team, epics are sequenced so that each sprint ends with a demonstrable, end-to-end increment (working data flow, workflow, or dashboard) rather than partially built components across many epics at once.

5.2 Roles

Role	Who	Responsibilities
Product Owner	Titan Technology Corp.	Owns and prioritizes the backlog; accepts completed stories at Sprint Review; is the escalation point for scope and business-rule decisions (approval routes, KPI definitions, branding assets).
Project Manager / Scrum Master	Titan Technology Corp.	Facilitates ceremonies, maintains the backlog and sprint schedule, manages the RAID log and stakeholder communication, removes blockers, tracks velocity and re-plans when scope/capacity shifts.
Power Platform Team	Titan Technology Corp.	Builds Power Apps forms (checklists, offline readings, FM intake), Power Automate flows (approvals, notifications, escalation), Dataverse schema, and the FM Portal (Power Pages).
Data Engineer / BI Team	Titan Technology Corp.	Builds the Medallion pipelines (Bronze/Silver/Gold), data models, Power BI semantic models and dashboards, and performance tuning.
Integration/Ops Team (Python/C#)	Titan Technology Corp.	Build custom connectors for BMS/PME where no native connector exists; supports peer testing across sprints. With a 3-person team, this role is absorbed by the Data Engineer with extended timelines on integration-heavy sprints (flagged in Section 7).

5.3 Ceremonies

Ceremony	Cadence	Purpose
Sprint Planning	Day 1 of sprint (2h)	Select and size stories for the sprint from the top of the backlog; confirm the sprint goal.
Daily Stand-up	Daily / 3x week, 15 min	Given the small team, a lightweight async or short sync check-in on progress and blockers.
Backlog Refinement	Mid-sprint (1h)	Product Owner and team clarify and estimate upcoming stories 1–2 sprints ahead.
Sprint Review / Demo	Last day of sprint (1h)	Demonstrate the working increment to stakeholders; gather feedback; formally accept stories.
Sprint Retrospective	Last day of sprint (30–45 min)	Team reflects on what worked/didn't and agrees on one or two improvements for the next sprint.

5.4 Definition of Ready (DoR)

- Acceptance criteria are written and understood by the team.
- Required data-source access/credentials (BMS, PME, other systems) are confirmed available, or a workaround/mock is agreed.
- UI stories have a wireframe/mock-up or reference screen agreed with the Product Owner.
- Dependencies on other stories/epics are identified and sequenced.

5.5 Definition of Done (DoD)

- Built component (flow, form, pipeline, dashboard) deployed to the Test environment and peer reviewed.

- Acceptance criteria verified and demoed; Product Owner has accepted the story.
- Data-quality checks (where applicable) pass with no unresolved critical issues.
- User-facing documentation updated (even briefly) and the RAID log updated if new risks/issues surfaced.

5.6 Release & Sprint Roadmap (indicative — 12 sprints, ~6 months)

Dates assume a kickoff on 7 September 2026 and exclude adjustment for public holidays; see Section 7 for the holiday-capacity assumption. ^(*) The full backlog behind each sprint is in the companion workbook's "Sprint Schedule" and "Product Backlog" tabs. Sprints are grouped below under the four milestone phases from the Executive Summary.

Phase 1 — Environment Access & Foundation

Spr.	Dates	Epic(s)	Sprint Goal / Key Deliverables
S1	07–18 Sep 2026	EPIC-0	Environment & architecture ready: Power Platform (Dataverse) and Fabric (OneLake) environments provisioned, Link to Microsoft Fabric configured, Medallion data model defined, source-system access confirmed, backlog refined.

Phase 2 — Automated Utilities Data & Data Central

Spr.	Dates	Epic(s)	Sprint Goal / Key Deliverables
S2	21 Sep–02 Oct	EPIC-1	Water data flowing reliably: BMS water connector writes into Dataverse (Data Central); Link to Fabric auto-syncs to Bronze; Silver cleansing & validation rules built.
S3	05–16 Oct	EPIC-1, EPIC-2	Power/PQM ingestion into Dataverse (Data Central), auto-syncing to Bronze → Silver; Gold-layer rollups (hourly/daily/monthly) for both water and power — the Fabric Medallion Lakehouse becomes operational.
S4	19–30 Oct	EPIC-1, EPIC-2	Utilities Power BI dashboards (Direct Lake over Gold) live in UAT: trends, campus/building/meter comparison, baseline/target views.

Phase 3 — Functional Workflows & Power BI Hub

Spr.	Dates	Epic(s)	Sprint Goal / Key Deliverables
S5	02–13 Nov	EPIC-3	Reusable workflow engine (Dataverse + Power Automate) built; CIWG/continuous-improvement workflow live end-to-end.
S6	16–27 Nov	EPIC-3	Risk register and FM projects/initiatives workflows configured on the shared engine; escalation rules active for all three FM workflows.
S7	30 Nov–11 Dec	EPIC-4	Digital checklist app (mandatory/range validation, timestamp/user ID) and critical-parameter trending dashboards live for pilot checklist types.
S8	14–25 Dec	EPIC-5	Offline reading-entry forms (D1-CAMPUS water meters, NVL line, other offline meters) live; manual readings integrated and compared against BMS/PME data. (spans year-end holidays — see Section 7)
S9	28 Dec–08 Jan	EPIC-6	KPI dashboards, part 1: CAPEX plan vs. actual, OHS/safety & compliance, equipment register/calibration status.
S10	11–22 Jan 2027	EPIC-6	KPI dashboards, part 2: training & skills, overtime tracking, vendor KPIs, incidents/near-misses/awards.

Spr.	Dates	Epic(s)	Sprint Goal / Key Deliverables
S11	25 Jan–05 Feb	EPIC-7	Existing dashboards/reports reviewed and consolidated; FM Portal / Power BI Hub built with operation branding, standard filters, and <3s performance tuning.

Phase 4 — Final Deliverable & User Training

Spr.	Dates	Epic(s)	Sprint Goal / Key Deliverables
S12	08–19 Feb	EPIC-8	End-to-end UAT and defect fixing, user/admin documentation, training sessions, post-go-live roadmap finalized, go-live and project closure.

6. Further Automation Opportunities — Post-Go-Live Roadmap

Once the platform in Sections 3–6 is live, the Medallion Gold layer and Dataverse become a foundation for higher-value automation. These are proposed as candidate initiatives for two post-go-live tracks, to be prioritized with FM once this project (Phases 1-4) is delivered and adopted.

6.1 Near-Term Track (approx. 3–6 months after go-live)

- **Anomaly detection & alerting:** apply threshold/statistical anomaly detection on Gold-layer water, power and PQM data (e.g. leak detection from sustained abnormal water flow, THD spikes) with proactive Power Automate alerts, extending FR-2.7.
- **Predictive maintenance:** use equipment-register + checklist + calibration history to flag assets trending toward failure or due for preventive maintenance, ahead of a hard due date.
- **Expand offline capture:** bring remaining manual meters/readings (beyond D1-CAMPUS and NVL) onto the same Power Apps pattern; evaluate low-cost IoT retrofits for the highest-value offline meters to move them into automated collection (EPIC-1/2 pattern).
- **Vendor & contract automation:** extend workflow automation (EPIC-3 engine) to vendor onboarding, contract renewal reminders, and automated vendor-KPI scorecards feeding Section 6 dashboards directly from SLA data.
- **Mobile push notifications:** move from email-only notifications (FR-3.3) to a dedicated mobile experience for technicians (approvals, overdue checklist reminders) via the Power Apps mobile app.

6.2 Longer-Term Track (approx. 6–12 months after go-live)

- **Conversational / chatbot interface:** a Copilot Studio (or equivalent) assistant over the FM Portal for status queries (“what’s the water usage trend for Building X this month”, “what CIWG requests are overdue”), reducing report-pulling effort.
- **Cross-campus benchmarking & sustainability reporting:** use the multi-year Gold-layer history (NFR-4) to support ESG/sustainability reporting and cross-campus efficiency benchmarking.
- **Advanced analytics / ML:** move selected Gold-layer workloads into a Fabric/Databricks notebook environment for load-forecasting, optimization models, or clustering of similar buildings for targeted efficiency programs.
- **CMMS/ERP integration:** if client adopts or already has a formal CMMS/ERP, integrate it with the FM Portal/Dataverse rather than duplicating equipment/finance data entry.

Each item below should be scoped and estimated (including a labor cost breakdown, intentionally excluded from this document) once it is prioritized, using the same epic → user story → story point structure established in the companion workbook.

Appendix A — Glossary

Term	Definition
Dataverse	Microsoft's low-code data platform underlying Power Apps/Power Automate. In this solution it is “Data Central” — the single system of record for both utilities readings and FM-process entities.
Microsoft Fabric / OneLake	Microsoft's unified analytics platform; OneLake is its single, tenant-wide data lake. The Medallion Lakehouse (Bronze/Silver/Gold) in this solution is built in Fabric on top of OneLake.
Link to Microsoft Fabric	A native Dataverse feature that continuously mirrors Dataverse tables into OneLake as managed Delta tables, with no custom ETL — used here to form the Bronze layer directly from Dataverse.
Direct Lake	A Power BI storage mode that reads Delta tables in OneLake directly, without importing or a separate refresh schedule — used here for the utilities and KPI dashboards over the Gold layer.
BMS	Building Management System — monitors/controls building mechanical & electrical equipment, incl. water/power metering.
PME / PQM	Power Monitoring Expert / Power Quality Monitoring — systems and metrics (voltage, THD, etc.) describing electrical power quality.
THD	Total Harmonic Distortion — a power-quality metric.
CIWG	Continuous Improvement Working Group — the SRS's term for continuous-improvement/change-request processes.
CAPEX	Capital Expenditure.
OHS	Occupational Health & Safety.
D1-CAMPUS	District 1 campus.
NVL	NVL line — a metering line referenced in the SRS for offline meters.
KPI	Key Performance Indicator.
ETL	Extract, Transform, Load.
ACID	Atomicity, Consistency, Isolation, Durability — transactional guarantees provided by the lakehouse storage layer.
DLP	Data Loss Prevention — Power Platform policies controlling which connectors can be combined.
ALM	Application Lifecycle Management.
MoSCoW	Prioritization scale: Must have, Should have, Could have, Won't have (this time).
RAID	Risks, Assumptions, Issues, Dependencies log.